Design of Supramolecular Nanofibers for Gonorrhea Vaccines

Prerana Kulla, Emily F. Roe, and Joel H. Collier

Department of Biomedical Engineering, Duke University, Durham, NC

Abstract

- Gonorrhea is a sexually transmitted disease (STD) that globally affects over 82.4 million people [1]
- Peptide vaccines, can be tailored to a variety of diseases by selecting the appropriate epitope, such as AniA - GALGQLKVEGAEN and TbpA2 - TRQICL for gonorrhea
- Peptide synthesis and purification are done with HPLC, LB, and MALDI
- TEM showed that both TbpA2-Coil29 and AniA-Coil29 produced nanofibers
- Zeta Protential showed that TbpA2-Coil29 had a negative charge and AniA-Coil29 had a positive charge
- AniA-Coil29 had a mucoadhesive level at 78.812. TpbA2-Coil29 had a mucoadhesive level at 20.225.

Background

To improve world health, vaccination must be widely practiced. Increased vaccination rates in lower- and middle-income nations face significant obstacles, such as the high costs associated with cold-chain delivery and the need for qualified medical personnel to give the vaccines. To overcome these challenges, we will design a heat-stable vaccination tablet that is administered via simple dissolving beneath the tongue. Importantly, these vaccinations can stimulate the production of protective antibodies in mucosal compartments, which can neutralize and eliminate infections from the body before they can establish themselves. Furthermore, they may offer considerable logistical and monetary advantages depending on the chosen mucosal delivery route.[2]

Nanofiber Platform

Nanofibers are fibers with a diameter on the nanometer scale and with a high aspect ratio [Fig 1.]. Self-assembling nanofibers can be produced from a variety of peptides, giving them a range of physical characteristics and possible uses. The nanofibers then self-assemble in physiological salt conditions.

High aspect ratio nanofibers made entirely of α-helical structure are created by the peptide Coil 29 [Fig 2.]. The fiber-forming peptide has its C-termini facing inward towards the fiber axis and their N-termini projecting radially from the fiber axis. Each layer in this structure is made up of four peptides [Fig. 3]. Peptide epitopes are conjugated to the N-termini on the outside which is why an immune response is produced. [3]

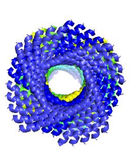


Fig 1.

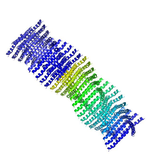


Fig 2.

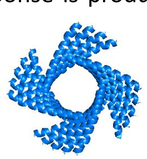


Fig 3.

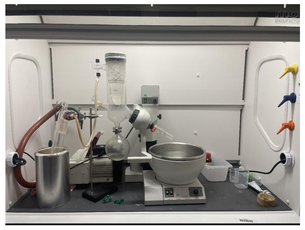
Peptide Synthesis and Cleavage

Peptide synthesis uses a Liberty Blue microwave-assisted peptide synthesizer which takes a programmed sequence of amino acids and attaches it to a resin bead that then repeats this step to create crude peptide [Fig 4.].

Cleavage, the next step, uses TFA, a strong acid, in order to remove the resin bead from the peptide. The rotovap is used to reduce the boiling point of the TFA to separate it from the peptide [Fig 5.].

Fig 4.Liberty Blue

Fig 5. Rotovap



Peptide Purification

Through the use of HPLC, peptide purification was accomplished [Fig 6.]. Samples of AniA-Coil29 and TbpA2-Coil29 were pipetted onto MALDI plates, which were then brought to the MALDI Mass Spectrometry system.

Fig 6. HPLC

A laser is used in MALDI to hit a matrix of tiny molecules. By calculating the time needed for an ion to transit the length of the flight tube, the molecular weight may be calculated. [4] AniA-Coil29 has a molecular weight of 5132Da [Fig 7.] and TpbA2-Coil29 has a molecular weight of 4585Da [Fig 8.], which both match expected mass values for synthesized peptides.

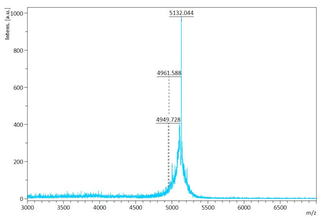


Fig 7. AniA-Coil29

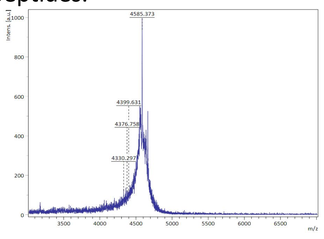


Fig 8.TbpA2-Coil29

Nanofiber Formation

The transmission electron microscopes (TEM) uses an electron particle beam to see specimens and produce a picture that is greatly enlarged. Both AniA- Coil29 [Fig 9.] and TbpA2- Coil29 [Fig 10.] produced nanofibers.

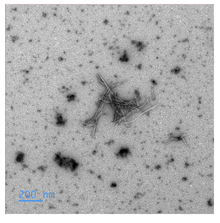


Fig 10.TbpA2-Coil29

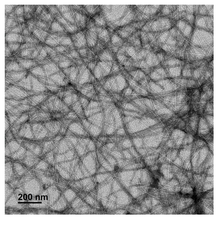
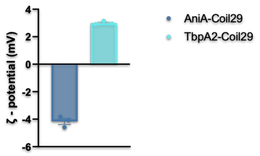


Fig 9. AniA-Coil29

Nanofiber Charge

Zeta potential is the charge that forms at the boundary between a solid surface and its liquid medium. Using the Zeta Potential machine, the charges of the nanofibers TbpA2-Coil29 and AniA-Coil29 were found. The first nanofiber, AniA-Coil29 had a mean Zeta Potential of -4.08 mV. The second nano fiber, TbpA2-Coil29 had a mean Zeta Potential of 3.15 mV. TbpA2-Coil29 has a negative charge while AniA-Coil29 has a positive charge.

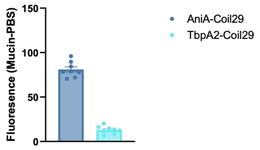
Fig 11.



Nanofiber Mucus Adhesion

With the Nanofiber Mucus Adhesion process, a lower signal corresponds to lower mucoadhesive levels and a higher signal corresponds to higher mucoadhesive levels. We can see that AniA-Coil29 has higher mucuhesive levels at around 78.812. TpbA2-Coil29 on the other hand has lower mucoadhesive levels at around 20.225 [Fig 12.]. This means that AniA-Coil29 has a harder time with penetration.

Fig 12.



Conclusions and Future Experiments

- Both AniA-Coil29 and TbpA2-Coil29 formed nanofibers
- Purified AniA-Coil29 had a molecular weight of 5132Da
- Purified Tbp2-Coil29 had a molecular weight of 4585Da
- AniA-Coil29 has a mean zeta potential of -4.08 mV
- TbpA2-Coil29 has a mean zeta potential of 3.15 mV
- AniA-Coil29 has mucoadhesive levels at 78.8
- TpbA2-Coil29 has mucoadhesive levels at 20.2

Future work will include testing the sublingual vaccine on mice. Every two weeks, 8 µl of 5 mM peptides will be administered sublingually to groups of 5 female C57BL/6 mice.

References

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Acknowledgements

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